MIL-P-45361B
20 November 1985
SUPERSEDING
MIL-P-45361A
5 February 1963

MILITARY SPECIFICATION

PUMP, FUEL, CAM-ACTUATED: ENGINE, MECHANICAL DIAPHRAGM AND PARTS KIT, ENGINE FUEL PUMP

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical fuel pumps of the cam-actuated, diaphragm type used on carburetor-fed internal combustion engines. This specification also covers parts kit for rebuilding fuel pumps of the type specified herein (see 6.2). The rated flow of the pumps covered herein ranges from 10 to 50 gallons per hour (gph) (see 3.3).

1.2 Classification. Items shall be classified as follows:

Class 1	Pump assembly.
Class 2	Parts kit assembled to pump.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2910

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

P-D-680	- Drycleaning Solvent.
VV-G-1690	- Gasoline, Automotive, Leaded or Unleaded.

MILITARY

MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-F-13927	- Fungus Resistance Test, Automotive Components.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Marking for Shipment and Storage.
MIL-STD-202	- Electronic and Electrical Component Parts, Test Methods for Electronic and Electrical Component Parts.
MIL-STD-45662	- Calibration Systems Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish pump or parts kit which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent pumps or parts kit delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

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3.2 Materials. Materials not designated on applicable drawings and not specifically covered by this or referenced specifications shall be uniform in quality and free from imperfections or defects which affect the serviceability of the pump or parts kit (see 4.8.1).

3.3 Design and construction. The design and construction of the pump and the parts kit shall be such as to assure performance in accordance with the requirements specified herein. Dimensions, rated flow, and the shut off pressures shall be in accordance with the applicable drawing or as specified in the contract or order (see 4.8.1 and 4.8.2).

3.3.1 Parts kit. The parts kit shall be comprised of all components specified on applicable drawings. When no drawings are referenced, the parts kit shall include all parts needed to rebuild the specified pump to the extent necessary to assure performance in accordance with the requirements specified herein (see 4.8.1 and 4.8.2).

3.3.2 Interchangeability. The parts kit, as a unit, shall be completely interchangeable with counterparts of the original pump (see 4.8.1).

3.4 Performance. The pump or parts kit assembled into a pump shall be capable of meeting all performance requirements specified herein while being operated under conditions simulating intended use.

3.4.1 Rated flow. The pump shall be capable of meeting the flow requirements at the respective outlet pressures specified in table I, while operated at a speed of 1800 cycles per minute (cpm) (see 4.8.3 and 6.3).

TABLE I. Flow rates.

Outlet pressure (percent of shut off pressure)	Percent rated flow
25	200
50	150
75	100

3.4.2 Shut off pressure. When the pump is operated at any speed from 200 to 1800 cpm, the shut off pressure at the pump outlet shall be as specified in the contract or order (see 4.8.4 and 6.2).

3.4.3 Endurance. The pump shall be capable of withstanding 500 hours of operation without failure, while mounted at an angle 30 degrees to the right or the left of the normal vertical operating position, as viewed in the cam shaft axial direction and while being subjected to the vibration resistance (see 4.8.5).

3.4.4 Leakage. The pump shall be capable of withstanding an internal air pressure of 10 pounds per square inch (psi) without any damage or leakage (see 4.8.6).

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3.4.5 Environmental conditions.

3.4.5.1 High temperature. The pump shall meet the performance requirements of 3.4.1 through 3.4.4, while operating in an ambient temperature of 160 degrees Fahrenheit (°F) (see 4.8.7.1).

3.4.5.2 Low temperature. The pump shall meet the performance requirements of 3.4.1 through 3.4.4, while operating in an ambient temperature of minus 25 ± 5°F, after being conditioned in ambient air at minus 65 ± 5°F. The rated flow due to this operation is as follows (see 4.8.7.2):

a. The pump shall be capable of priming itself and delivering not less than two fluid ounces of fuel within the first 120 cycles of operation, while operating at a speed of 200 cpm with a rated flow of 10 gph or less.

b. The pump shall be capable of priming itself and delivering not less than two fluid ounces of fuel within the first 40 cycles of operation, while operating at a speed of 200 cpm with rated flow of 10 gph or more.

c. The pump shall be capable of meeting the performance requirements of 3.4.2 while being operated at 200 cpm and under the conditions specified in a and b.

3.4.5.3 Fungus resistance. The pump shall meet the performance requirements of 3.4.1 through 3.4.4, after being inoculated and exposed to a fungus supporting environment as specified in MIL-F-13927 (see 4.8.7.3).

3.5 Marking. Pumps and parts kit shall be marked in accordance with MIL-STD-130 (see 4.6.2).

3.5.1 Part designation. The inlet or outlet part of the pump shall be clearly and permanently designated on the casting.

3.6 Workmanship. Workmanship shall be such as to assure that pumps and parts kits are free of burrs, scratches, rough edges, chips, rust, foreign matter of any kind, or unnecessary grease (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 $\pm$ 10°C
b. Barometric pressure 725 + 50mm Hg
   - 75
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select pumps or parts kit produced under the production contract (see 6.2) for first article inspection. First article samples shall be inspected in the sequence listed in table II and as specified in table III. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply pumps or parts kit that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

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TABLE II. First article testing.

Sample No.	Test	Paragraph
1	Interchangeability test (when applicable) Leakage test Flow rate test Shut off pressure test High temperature test Low temperature test Endurance test Flow rate test Shut off pressure test Leakage test	4.8.1 4.8.6 4.8.3 4.8.4 4.8.7.1 4.8.7.2 4.8.5 4.8.3 4.8.4 4.8.6
2	Interchangeability test (when applicable) Leakage test Shut off pressure test Flow rate test Fungus resistance test Flow rate test Leakage test Shut off pressure test	4.8.1 4.8.6 4.8.4 4.8.3 4.8.7.3 4.8.3 4.8.6 4.8.4
3	To be used in case sample No. 1 or 2 is damaged inadvertantly.	

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Acceptance (100%)	
Materials and construction	3.2 thru 3.3.2	4.8.1	X			
Defects	3.3, 3.3.1, 3.5, & 3.6	4.8.2	X	X		
Rated flow	3.4.1	4.8.3	X		X	
Shut off pressure	3.4.2	4.8.4	X		X	
Endurance	3.4.3	4.8.5	X			X
Leakage	3.4.4	4.8.6	X			X
High temperature	3.4.5.1	4.8.7.1	X			X
Low temperature	3.4.5.2	4.8.7.2	X			X
Fungus resistance	3.4.5.3	4.8.7.3	X			

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4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the pumps or parts kits of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1 and 3.3.2).	SIE 1/
103	Identification marking, improper (see 3.5).	Visual
104	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1 and 3.3.2).	SIE 1/
202	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Acceptance tests (100 percent). Each pump or part kit shall be subjected to the acceptance tests specified in table III.

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4.6 Control tests. When specified (see 6.2), control tests shall be conducted on two from each lot of 500 units consecutively produced, except that not more than two tests shall be performed in a one month period. The pumps or parts kits units shall be subjected to the control tests specified in table V.

TABLE V. Control tests.

Sample No.	Test	Paragraph
1	Flow rate test Low temperature test Flow rate test	4.8.3 4.8.7.2 4.8.3
2	Flow rate test Endurance test Flow rate test	4.8.3 4.8.5 4.8.3

4.7 Failure. Failure of any pump or parts kit to pass any of the specified acceptance shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.3.1, 3.5 and 3.6 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Flow rate. To determine conformance to 3.4.1, the pump shall be mounted on a test fixture and operated for five minutes at each of the outlet pressures specified in table I. The flow rate shall be measured during each of the five minute period. If the flow rate at any specified pressure is less than the values specified in table I, the pump shall be rejected.

4.8.4 Shut off pressure. To determine conformance to 3.4.2, the pump shall be mounted on a test fixture and operated with no flow restriction for a period of one minute. The flow shall then be restricted to zero. After an interval of one minute, the shut off pressure shall be recorded at the pump outlet. If the shut off pressure is not as specified in the contract or the order, the pump shall be rejected.

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4.8.5 Endurance test. To determine conformance to 3.4.3, the pump shall be mounted on a test fixture capable of being adjusted to provide pump operation at any angle from 30 degrees to the right to 30 degrees to the left of the normal vertical operating position, as viewed in the cam shaft axial direction. The test fixture shall be mounted on a vibrating machine capable of providing vibration in accordance with MIL-STD-202, method 201A during the entire period comprising the endurance test. Pump position, flow rate and pressure factors during the endurance test shall be in accordance with table VI.

TABLE VI. Endurance test factors.

Test hours	Pump position			Rated flow (percent)	Outlet pressure (percent minimum shut off pressure)
	30 degrees right	Vertical	30 degrees left		
95		X		10	
1		X			75
95		X		20	
1		X			75
95		X		30	
1		X			75
95		X		40	
1		X			75
95		X		50	
1	X				75
6			X		75
6		X			75

4.8.6 Leakage. To determine conformance to 3.4.4, the pump shall be subjected to 10 psi internal air pressure while the fuel section is submerged in water and the outlet port is plugged. The pump shall be rejected if there is any leaks other than the inlet connections or the outlet plug.

4.8.7 Environmental conditions.

4.8.7.1 High temperature. To determine conformance to 3.4.5.1, the pump shall be filled with solvent (stoddard) conforming to type II of P-D-680, and shall be conditioned for not less than 4 hours in an ambient temperature of 160°F. The pump shall pass the test of 4.8.3 while operating in the same ambient temperature except that the fluid test temperature shall not exceed 100°F.

4.8.7.2 Low temperature. To determine conformance to 3.4.5.2, the test fluid shall be gasoline conforming to type II of MIL-G-3056. The test fluid and the pump shall conditioned in an ambient temperature of minus 65 $\pm$ 5°F for not less than 16 hours. The test shall be conducted with the test fluid and the pump at minus 25 $\pm$ 5°F. The pump shall be drained prior to the test and shall be operated during the test at a speed of 200 cpm. The test results shall be evaluated as follows:

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a. Pumps with rated flow of 10 gph or less shall be operated at the specified speed for a total 120 cycles. If the pump fails to deliver at least 2 fluid ounces during the period of operation, the pump shall be rejected (see 3.4.5.2).
b. Pumps with rated flow of more than 10 gph shall be operated at the specified speed for a total of 40 cycles. If the pump fails to deliver at least 2 fluid ounces during the period operation, the pump shall be rejected (see 3.4.5.2).
c. Following the above, the pump shall be tested as specified in 4.5.2.2, except that temperature range and operating speed shall be as specified above.

4.8.7.3 Fungus resistance test. To determine conformance to 3.4.5.3, the pump shall be tested in accordance with class 2, method A of MIL-F-13927, except that specimens shall be incubated for one continuous period of 90 days.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. Pumps covered by this specification are intended for use in fuel supply systems and installation on internal combustion gasoline engines and are intended for use with fuels conforming to MIL-G-3056 or VV-G-1690. This type pump has a reciprocating action with a positive suction stroke, and a discharge stroke which is powered by a calibrated spring.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specify whether acquisition is for pumps or for parts kits (see 1.1).
c. If first article samples are not required (see 3.1).
d. Number and date of the applicable drawing (see 3.3).
e. Rated flow and shut off pressure, when applicable (see 3.4.1 and 3.4.2).
f. If responsibility for inspection shall be other than as specified (see 4.1).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
h. If inspection conditions shall be other than as specified (see 4.3).
i. If first article inspection is required (see 4.4).
j. Whether control tests are applicable (see 4.6).
k. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

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6.3 Definitions.

6.3.1 Rated flow (see 3.4.1). Pump delivery in gallons per hour (gph), as specified in the contract or order.

6.3.2 Shut off pressure (see 3.4.2). The term "shut off pressure", as used herein, means the pressure in pounds per square inch recorded at the pump outlet while the pump is in operation and while flow is restricted to zero.

Custodians:
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Navy - YD
Air Force - 99

Review activities:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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